ZHIPAN XU

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EDUCATION

University of California, Berkeley **Aug. 2026 – May 2027**
Master of Engineering in Electrical Engineering and Computer Sciences
Honors: Fung Excellence Award

Nanjing University **Sep. 2021 – Jun. 2025**
Bachelor of Science in Computer Science and Technology | GPA: **4.45/5.0**
Honors: First-Class Scholarship (Top 15%), 3 times

PUBLICATIONS

Zhipan Xu and Lijun Zhang. *Online Non-convex Learning in Dynamic Environments*. **NeurIPS 2024**.

PROFESSIONAL EXPERIENCE

EvoQuant Technology **May 2025 – Aug 2026**
Quantitative Researcher *Shenzhen, China*

Deep Learning for Cross-Commodity Futures Forecasting

- Trained and tested multiple deep learning models, including TabM, GRU, and Transformers, to find predictive signals not captured by tree-based models.
- Developed an ensemble framework using random feature subsets, improving rolling out-of-sample information coefficient (IC) by **23%** over a single tabular model.
- Combined selected deep learning models with LightGBM in a live trading ensemble, improving out-of-sample IC by **38%** over the baseline and contributing **positive live P&L**.

- **Signal Evaluation and Loss Design:** Developed a metric that accounts for turnover and transaction costs to compare futures signals across prediction horizons. Used it to select prediction targets and design turnover-regularized losses.
- **Learnable Feature Aggregation:** Designed a learnable layer to share features across instruments. Applied hierarchical clustering to the learned weights across **100+** commodity futures to guide the design of training sample weights.
- **Research Data Pipeline:** Reduced data preparation time by **80%** by redesigning time-series indexing and sample generation for approximately 200 million records and 5,000 features.

X-Square Asset **Feb. 2025 – May 2025**
Quantitative Research Intern *Shanghai, China*

- **Feature Selection:** Designed an iterative feature-selection and reweighting framework for linear trading models, improving validation performance while reducing the feature set by **10%** across seven futures contracts.
- **Nonlinear Return Prediction:** Developed a LightGBM model for minute-level return prediction on CSI 1000 index futures, improving R^2 by **0.002** and backtested net returns by **12%** over a linear regression baseline.
- **Online Strategy Selection:** Developed an online strategy-selection system with daily selection from a candidate pool maintained using market volatility and recent performance; improved average backtested net returns by approximately **20%** over the best fixed strategy across 10 futures products.

RESEARCH EXPERIENCE

Berkeley Artificial Intelligence Research (BAIR) **Aug. 2026 – Present**
Graduate Student Researcher *Berkeley, CA*

- Developing safe robotic manipulation policies with reinforcement learning to improve task success and reduce runtime safeguard costs. Built a custom MJX-Warp environment for Brax PPO training and diffusion policy fine-tuning with DPPO.

LAMDA Group, Nanjing University **Jul. 2023 – Mar. 2024**
Undergraduate Researcher *Nanjing, China*

- Proposed a two-layer expert/meta-algorithm framework with randomized sampling for online learning with non-convex losses and time-varying data distributions; designed dynamic-regret and adaptive-regret algorithms and **proved optimal-order regret bounds**.
- Published the work as first author at **NeurIPS 2024**. Validated the algorithms on online constrained meta-learning, reducing cumulative loss by 34.80% over static baselines in imitation learning and improving robust 5-shot CUB image classification accuracy by 9.93 percentage points.

OPEN-SOURCE PROJECT

PyNOL: Python Toolkit for Non-Stationary Online Learning [GitHub] **Dec. 2023 – May 2024**

- Extended PyNOL with FTPL and Adapt-ML-Prod algorithms, adding modular model updates and randomized mixtures. Enabled evaluation of online algorithms in batch-learning settings.

TECHNICAL SKILLS

Programming: Python, C, C++, SQL, MATLAB, Java
Mathematics: Probability Theory, Mathematical Statistics, Stochastic Processes, Linear Algebra, Optimization
Machine Learning & Statistics: PyTorch, JAX, scikit-learn, Pandas, NumPy, Xarray
Tools & Systems: Git, Linux, Shell, Docker, MuJoCo (MJX), Hadoop MapReduce, LaTeX

AWARDS

First Prize (Top 10%), National Olympiad in Informatics in Provinces (NOIP) **2019, 2020**